

Reliable Latent-Tree Clustering under Static and Drifting Distributions

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- 4 模板 (最后再删, 用于粘贴)

An example of a latent tree

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分两类时的两种分法

- Can latent variables extract information more conducive to clustering while reducing the number of variables (possibly enabling more categories)?

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Exploring the Impact of Latent Variables on Clustering!

How to conduct it

放一个图片，第一个是数据，指向图，指向 H 的数据，指向聚类，同时 O ， $O+H$ ，等都指向聚类，最后再写放一个漂移项。

1. 结构学习和参数估计（17 年 20 年的工具）
2. 聚类（如何利用上一步的信息进行聚类）
3. 如果数据是具有漂移分布的

结构学习参数估计

只放最简单的如何结构学习和参数估计

Types of data representations

- **Observed-only** (O)—the raw observed data, serving as the standard input for most clustering methods.
- **Latent-only** (H)—probabilistic distributions of latent variables inferred from observed data.
- **Observed + Latent** (O+H)—a concatenation of observed data and latent variable distributions.
- **Latent + Subtree** (H+ T_H)—latent variable distributions enriched with their local subtree structure T_H using Graph Convolutional Networks (GCN).
- **Observed + Latent + Tree** (O+H+ T)—a fusion of observed and latent data further integrated with the entire latent tree structure T via GCN.

Clustering methods

- K-Means clustering
- Gaussian mixture models (GMM) clustering
- spectral clustering
- Sinkhorn-Knopp clustering
- Co-clustering
- Dual Mutual Information Constraint Clustering (DMICC)

ODE model of parameter evolution.

Suppose the true underlying parameter trajectory $\theta(t) \in \mathbb{R}^d$ evolves according to the following first-order ODE:

$$\frac{d\theta(t)}{dt} = -\nabla V(\theta(t)), \quad \theta(0) = \theta_0,$$

Exponential moving averages (EMA)

SLLT: updated estimates of the first- and second-order moments via EMA.

PEMT: higher-order moments are also updated via EMA.

Internal metrics

- Silhouette Coefficient (SI)
- Calinski-Harabasz Score (CH)
- Davies-Bouldin Index (DBI)↓

External metrics

- Adjusted Rand Index (ARI)
- Normalized Mutual Information (NMI)
- Clustering Accuracy (ACC):

Identifiability and Assumptions

Identifiability

- Structural Identifiability: $p(x, T_1) = p(x, T_2) \Rightarrow T_1 = T_2$
- Parameter Identifiability: $p(x, \theta_1) = p(x, \theta_2) \Rightarrow \theta_1 = \theta_2$

Three standard assumptions

- (A1) Every pair of variables has a nonzero correlation.
- (A2) Each latent variable has at least three neighbors.
- (A3) Any two connected variables are not fully dependent.

These assumptions are standard assumptions (Pearl, 1988) to ensure that our learning graphical model is connected and the model doesn't include a redundant latent variable.

Theorem 1 (Identifiability of mixed latent tree models)

Assume three standard assumptions hold. Given the true first and second moments of the observed variables, the latent tree structure is identifiable. Given the moments up most to the fifth order, the model parameters of a mixed latent tree model are generically identifiable with probability one, up to label swapping.

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最优设计: 在指定的模型下, 有某种最优性的设计点.

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基于最优设计的子样本:

- 达到某些最优性质.
- 尽可能包含更多信息.
- 依赖模型.

Thanks!